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Roll No

EC-4003-CBGS

B.E., IV Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Integrated Circuits and its Applications

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Sketch neat diagram if any.

1. a) Explain Barkhausen criterion and compare positive and negative feedback. 7
b) Describe Wien bridge oscillator in detail. 7
2. a) What is single input balanced output differential amplifier? 7
b) Define the terms: 7
 - i) Slew rate
 - ii) Input bias current
 - iii) Gain bandwidth product
 - iv) Thermal drift
3. a) Describe Chebyshev filter and its characteristics. 7
b) Draw the waveform and circuit diagram of rectangular wave generator. 7
4. a) Explain Bistable multivibrator using IC-555 timer. 7
b) Explain Schmitt trigger with waveform. 7

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- 5. a) Explain op-amp voltage regulator in detail. 7
- b) What is dual power supply? Discuss basic switching regulator. 7
- 6. a) Describe the concept of positive feedback. 7
- b) Explain dual input unbalanced output differential amplifier. 7
- 7. a) Describe the integrators and differentiator circuit diagram. 7
- b) Discuss peak detector with waveform. 7
- 8. Write short notes on: 14
 - a) Voltage limiter
 - b) Crystal oscillator
 - c) Notch filter
